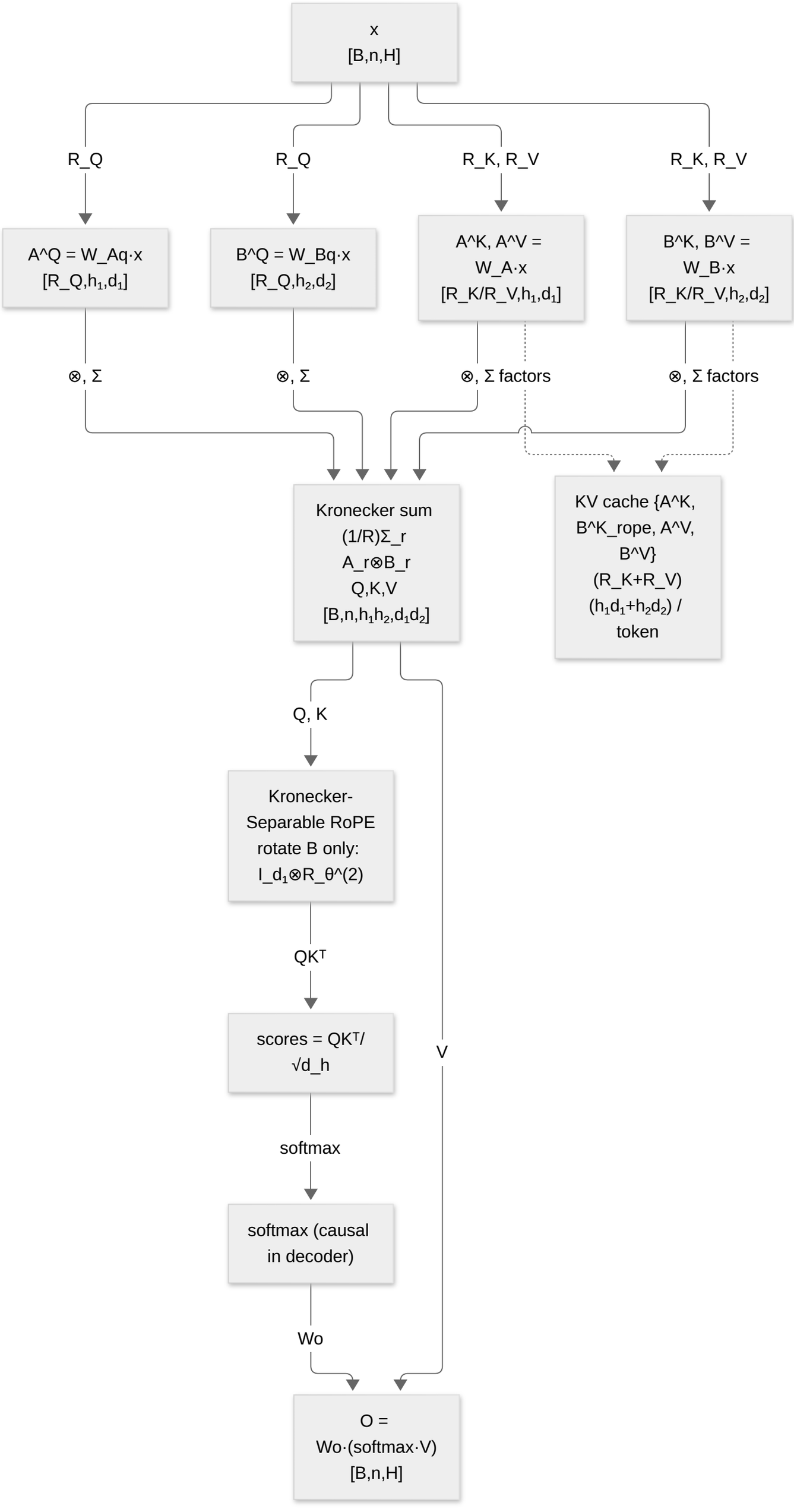

x
[B,n,H]
R_Q
R_Q
R_K, R_V
R_K, R_V
A^Q = W_Aq·x
[R_Q,h1,d1]
B^Q = W_Bq·x
[R_Q,h2,d2]
A^K, A^V =
W_A·x
[R_K/R_V,h1,d1]
B^K, B^V =
W_B·x
[R_K/R_V,h2,d2]
⊗, Σ
⊗, Σ
⊗, Σ factors
⊗, Σ factors
Kronecker sum
(1/R)Σ_r
A_r⊗B_r
Q,K,V
[B,n,h1h2,d1d2]
KV cache {A^K,
B^K_rope, A^V,
B^V}
(R_K+R_V)
(h1d1+h2d2) /
token
Q, K
Kronecker-
Separable RoPE
rotate B only:
I_d1⊗R_θ^(2)
QKᵀ
scores = QKᵀ/
√d_h
softmax
softmax (causal
in decoder)
Wo
V
O =
Wo·(softmax·V)
[B,n,H]